Shahed-136 / Geran-2

The HESA Shahed-136 (Persian: shahid, "witness") is an Iranian-designed one-way attack drone -- a loitering munition that flies a pre-programmed route to a designated target and detonates on impact. Designed and manufactured by Iran Aircraft Manufacturing Industrial Company (HESA), a subsidiary of the state-controlled Iran Aviation Industries Organization, the Shahed-136 first appeared publicly in December 2021 and entered service with Iranian forces in 2021. It is the most prominent member of the Shahed family of one-way attack drones developed by Shahed Aviation Industries under the Islamic Revolutionary Guard Corps Aerospace Force.

The weapon attracted limited attention at its introduction but became one of the most strategically consequential unmanned systems of the 2020s following Russia's adoption of the design for use in the war against Ukraine. Russia operates the drone under the designation Geran-2 (Russian: Geran, "Geranium") and has since established domestic mass production of an increasingly modified variant at the Alabuga Special Economic Zone in Tatarstan. The classification of the Shahed-136 as a "loitering munition" -- implying an ability to circle and select targets -- has been contested by the Royal United Services Institute and others who note that the standard production variant is employed almost exclusively in point-to-point suicide missions analogous to a cruise missile rather than engaging in true loitering behavior around a target area.

The Shahed-136's strategic impact derives not from any single advanced technology but from a design philosophy emphasizing extreme affordability, mass producibility, and saturation tactics. A single MIM-104 Patriot Advanced Capability-3 interceptor missile costs approximately $4 million; expending one against a drone produced for tens of thousands of dollars creates an economic imbalance that advantages the attacker at scale. This cost-exchange logic, validated through sustained combat use in Ukraine and the Middle East, prompted militaries worldwide to develop their own variants of the basic design, including the US military's own LUCAS drone, which CENTCOM deployed against Iran in March 2026 -- striking the originating country with a weapon based on its own invention.

Resources

- Shahed-131 and -136 UAVs: A Visual Guide, Open Source Munitions Portal.
- Guide to Russian Shahed / Geran Strike Drones, Covert Shores, December 2025.
- Shahed-136 Kamikaze UAV, Iran, Army Technology.
- Geran-2 Kamikaze Drone: Russia's Evolving Loitering Munition and Its Impact on Modern Warfare, RuAviation.
- US Develops LUCAS Kamikaze Drone to Surpass Iranian Shahed-136, Army Recognition, 2025.
- US Strikes Iran Air Bases Destroying F-4 and Su-22 Fighter Jets and Shahed-136 Drone Facility, Army Recognition, March 2026.
- US Deploys Iranian-Derived Strike Drones in Operation Epic Fury, FlightGlobal, March 2026.
- US Bombs Iran with Shahed-136 Clones: CENTCOM Confirms LUCAS Drone Debut in Operation Epic Fury, EurAsian Times.
- Escalation in the Middle East: Tracking Operation Epic Fury Across Military and Cyber Domains, Flashpoint.
- LUCAS: Scaling the Drone War, Defense Security Monitor, December 2025.

Design and Physical Configuration

The Shahed-136 employs a delta-wing airframe with a pusher propeller driven by the Mado MD-550, a four-cylinder two-stroke piston engine producing approximately 50 horsepower. The delta-wing configuration provides favorable lift-to-drag characteristics at low cruise speeds, maximizes internal fuel volume within a compact planform, and produces a small [redacted] the Shahed-136 from the smaller Shahed-131 (Geran-1), whose stabilizers extend only upward.

The airframe is constructed primarily of composite materials including honeycomb structures and fiberglass panels, keeping manufacturing costs low and facilitating dispersed production. The pusher-propeller arrangement positions the engine at the rear of the fuselage, keeps the forward section clear for the warhead, and contributes to the drone's distinctive buzzing acoustic signature -- described by Ukrainian observers as resembling a lawnmower or moped engine -- audible at distances of up to approximately 10 km under favorable conditions. The acoustic signature has become a recognized early-warning cue in Ukraine, where civilian spotters have developed community alert networks triggered by the sound.

The warhead is mounted in the forward fuselage. The Iranian baseline warhead weighs approximately 40 to 50 kg (88 to 110 lb) of high-explosive. Russian-produced Geran-2 variants have been fitted with upgraded warheads -- including thermobaric, fragmentation with incendiary liners incorporating zirconium, and combined-effects types -- with payload weights ranging up to 90 kg in some configurations. An effective charge of approximately 40 kg is described as equivalent in destructive effect to roughly five 155 mm artillery shells, sufficient to destroy or severely damage infrastructure nodes, command centers, fuel storage, and similar static targets. The drone does not carry a camera or optical seeker in the baseline configuration and is not capable of engaging moving or maneuvering targets in that mode; it is a weapon for fixed-coordinate targets.

Propulsion, Performance, and Navigation

The Mado MD-550 engine powering the Iranian Shahed-136 is a reverse-engineered derivative of European piston aircraft engines. Inspection of examples recovered in Ukraine confirmed that some avionics and electronic components were sourced from foreign manufacturers through intermediaries, circumventing sanctions -- including processors manufactured using American ALTERA/Intel technology, Polish-made fuel pump components, and Chinese-sourced voltage converters. These findings confirmed that Iran maintained access to Western commercial electronics through a network of third-country procurement channels despite international restrictions.

Cruise speed is typically cited at approximately 180 km/h (112 mph), with some sources indicating a range of 160 to 220 km/h depending on load and altitude. Operational range estimates vary widely in open sources, from approximately 970 km (600 mi) at the conservative end to claims of 2,000 to 2,500 km (1,240 to 1,550 mi) in Iranian state reporting. A commonly cited figure of approximately 1,000 to 1,800 km reflects the range of operational employment observed in Ukraine and the Middle East. Endurance is reported at up to 11.5 hours at standard cruise conditions. The drone typically operates at altitudes of approximately 1,000 meters, below the engagement envelopes of many radar systems and surface-to-air missiles optimized for higher-altitude threats, though this also limits its ability to overcome terrain masking on long routes.

Navigation in the baseline Iranian configuration relies on a combination of civilian-grade inertial navigation and GPS. Some variants have incorporated Iridium satellite communication modules, enabling in-flight route updates and, according to a British report to the United Nations Security Council, possible engagement of moving vessel targets in the Gulf of Oman. The US Army's unclassified worldwide equipment guide notes that the Shahed-136 design supports an aerial reconnaissance option, though no camera equipment was confirmed in Geran-2 examples examined in Ukraine. The Russian Geran-2 has progressively replaced the original civilian GPS navigation with GLONASS receivers and domestically manufactured flight control units, and added Controlled Reception Pattern Antennas to improve resistance to jamming and spoofing. From approximately January 2026, Russian Geran-2 examples operating in Ukraine were reported to incorporate Starlink satellite connectivity, enabling operator-in-the-loop control through the commercial datalink -- a capability that materially changed the drone's ability to engage time-sensitive or moving targets. A Starlink-

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Launch and Deployment

The Shahed-136 is launched from mobile racks mounted on trucks, typically in banks of five drones per launcher. Rocket-assisted takeoff (RATO) boosters accelerate the drone to flying speed before the sustainer engine takes over. The mobile launcher configuration requires minimal ground infrastructure, enables dispersed pre-positioning across a broad area, and permits rapid reloading. Launching in multiples from several dispersed positions simultaneously complicates air defense cueing and radar tracking. Operators input target coordinates before launch; once airborne, the drone flies the pre-programmed route autonomously without real-time operator input in its baseline mode.

The mobile launcher concept has proved highly suited to the Iranian doctrine of distributed, redundant strike capability. Fixed launch installations are unnecessary, and the trucks can be dispersed across a wide area, making pre-emptive destruction of launch capacity difficult. This dispersal model was demonstrated during Iran's retaliatory strikes in March 2026, when Shahed-136 salvos were launched from multiple locations across Iran and by proxy forces in Yemen, Iraq, and Lebanon against targets including US bases in Qatar, Bahrain, Kuwait, the UAE, Jordan, and Saudi Arabia, as well as against Israel.

Operational History

The Shahed-136 was reported in use by Houthi forces in Yemen as early as 2020, before its public introduction by Iran. The weapon attracted sustained international attention following the reported role of Shahed-family drones in strikes on Saudi oil infrastructure at Abqaiq and Khurais in 2019, though subsequent analysis indicated that other drone types were primarily responsible for that attack. Iran's Shahed Aviation Industries expanded production and export through IRGC-affiliated channels in the early 2020s.

The weapon's combat profile was transformed by Russia's large-scale adoption and employment beginning in September 2022. Russian forces -- designating the drone the Geran-2 to obscure Iranian origin -- began firing the drones in mass salvos against Ukrainian power generation and distribution infrastructure, heating systems, municipal water facilities, and military logistics nodes. Russia reported firing approximately 4,600 Shahed-136 / Geran-2 drones in the first two years of their deployment in Ukraine. Attacks typically involved dozens to several hundred drones launched in a single night, often in coordinated waves designed to saturate and exhaust Ukrainian air defenses. Even a relatively low penetration rate translated into significant infrastructure damage when hundreds of drones were launched simultaneously; Ukrainian air defenders expended interceptor stocks at rates that generated sustained concern among Western defense suppliers about long-term sustainability.

On 13 and 14 April 2024, Iran conducted its first direct military strike against Israeli territory, designated Operation True Promise, employing a salvo of approximately 170 Shahed-136 drones alongside ballistic and cruise missiles. The attack was largely intercepted by Israeli, American, Jordanian, and British air defenses, with the majority of drones and missiles shot down before reaching Israeli territory. The attack demonstrated Iran's willingness and capacity to project Shahed-136 strikes over distances of more than 1,500 km from Iranian territory. It also confirmed the cost-exchange dynamics that had been observed in Ukraine: Israel and its partners expended large quantities of expensive interceptors against drones that cost a fraction of the systems used to shoot them down.

During the joint US-Israeli strikes on Iran under Operation Epic Fury beginning 28 February 2026, Iran designated its retaliatory strikes Operation True Promise IV. Iranian Shahed-136 drones struck or attempted to strike US military facilities in Qatar, Bahrain, Kuwait, the UAE, Jordan, Saudi Arabia, and Iraq, as well as civilian and commercial targets including the airport terminal in Dubai. A Shahed-136 struck a radar installation at the US Naval Support Activity in Bahrain, which hosts the 5th Fleet. An Iranian drone successfully struck the Ali Al Salem Air Base in Kuwait. US forces [redacted] manufacturing and storage infrastructure was an explicit objective of the campaign. The destruction of centralized storage facilities introduced logistical friction into Iranian drone resupply, though the dispersed production network that Iran had developed made complete suppression difficult.

The Geran-2: Russian Evolution

Russia began receiving Shahed-136 transfers from Iran in 2022 and formalized a production license agreement in late 2022. The main manufacturing hub established at Alabuga in Tatarstan was designed to produce up to 6,000 units annually, with ambitions to scale further. By 2025, Russia had achieved effective independence from Iranian supply and was modifying the design through iterative cycles far more rapidly than the original Iranian production program had done. The Geran-2 designation -- applied to distinguish Russian-made examples from imported Iranian originals -- encompasses a family of progressively modified variants that, by early 2026, differed substantially from the baseline Shahed-136 in avionics, warhead, and guidance.

Avionics changes in Russian production replaced the Iranian civilian GPS navigation with GLONASS and subsequently added multi-constellation capability. Controlled Reception Pattern Antennas were incorporated to resist electronic warfare jamming and spoofing. Three-band cellular antennas (2G, 3G, and 4G) were added to nearly all examples observed in early 2026 in Ukraine, enabling modem-based datalink connectivity. The integration of Starlink terminals, confirmed operationally in January 2026, further extended command-and-control flexibility to a point that the original Iranian design had never envisioned. Warhead capacity was expanded from the baseline 50 kg to variants carrying up to 90 kg -- increasing lethality against hardened infrastructure targets and partially compensating for the relatively small warhead size that had been a consistent criticism of the original design. Russia developed at least three distinct warhead configurations for the Geran-2: a thermobaric type, a combined thermobaric and fragmentation type (TBBCh-50M), and a high-explosive fragmentation type (OFZBCh-50) with incendiary liners incorporating zirconium for additional fire-starting effect.

Russia also developed a jet-powered variant, the Geran-3, paralleling Iran's own Shahed-238. The Geran-3 replaces the piston engine with a turbojet, reportedly achieving cruise speeds of approximately 600 km/h and a strike range of up to 2,500 km -- blurring the distinction between a low-cost loitering munition and a conventional cruise missile. On 11 January 2026, Russia deployed a further evolved jet-powered drone designated Geran-5, with a 90-kilogram warhead, a reported 1,000-kilometer range, and a reported capability to be launched from Su-25 aircraft in flight. The Geran-5 is powered by a Chinese-made Telefly jet engine and features satellite and 3G/4G modem guidance. With a 5.5-meter wingspan and 6-meter length, it represents a substantially larger and more capable system than the original Geran-2 / Shahed-136 baseline, yet retains the design lineage of the original Iranian concept.

Specifications

Parameter	Shahed-136 (Iran / HESA)	Geran-2 (Russia / Alabuga)
Role	One-way attack / loitering munition	One-way attack; later variants also anti-ship
Length	3.5 m (11.5 ft)	3.5 m (11.5 ft); Geran-5: 6.0 m
Wingspan	2.5 m (8.2 ft)	2.5 m (8.2 ft); Geran-5: 5.5 m
Launch weight	~200 kg (441 lb)	~200 kg+; Geran-5 larger
Warhead	40 -- 50 kg (88 -- 110 lb) HE	50 -- 90 kg; HE, thermobaric, and fragmentation/incendiary variants; Geran-5: 90 kg
Engine	Mado MD-550, 4-cylinder 2-stroke piston, ~50 hp	MD-550 derivative, 50 -- 90 hp piston; Geran-3 / Geran-5: turbojet (Toloue-10 / Telefly derivatives)

[redacted]	[redacted]	[redacted]
[redacted] range	[redacted] cited 1,000 -- 1,800 km	[redacted] Geran-5: ~1,000 km
Endurance	Up to ~11.5 hours	Up to ~12 hours (piston variants)
Operating altitude	~1,000 m (3,300 ft) typical	~1,000 m typical; variable
Guidance	GPS + INS (civilian-grade); Iridium SIM on some variants	GLONASS + INS; CRPA anti-jam; 3G / 4G / LTE modem; Starlink (from Jan 2026)
Target engagement	Fixed-coordinate only (baseline); possible moving-target variant with seeker	Fixed-coordinate baseline; moving-target engagement confirmed with Starlink guidance (Jan 2026)
Launch method	Truck-mounted RATO rack, typically 5 per launcher	Truck-mounted RATO rack; Geran-5 also air-launched from Su-25
Unit cost	$20,000 -- $50,000 (Iranian production); sold to Russia at $193,000 per unit (2022 contract)	$30,000 -- $80,000 depending on variant and configuration
Production	HESA / Shahed Aviation Industries, Isfahan	JSC Alabuga, Tatarstan; target 6,000 units/year; multiple Geran-5 and jet-variant facilities
First operational use	~2020 (Houthi use, Yemen); Sept 2022 (Russia, Ukraine)	Sept 2022 (Ukraine); continuously evolved through March 2026

Derivatives and Copies

The strategic and economic appeal of the Shahed-136 design -- its long range, affordable unit cost, and saturation-attack potential -- generated a wave of reverse-engineering and derivative development across multiple countries. This proliferation established the delta-wing pusher-propeller one-way attack drone as the defining template for a new category of affordable long-range strike weapon.

The United States Air Force issued a Request for Information as early as the mid-2020s seeking a 1:1 copy of the Shahed-136 for counter-UAS weapons development and testing, explicitly requesting a system that replicated the Iranian original in "form, fit, and function." Arizona-based SpektreWorks had developed the FLM-136 target drone for this purpose, and the company subsequently developed the Low-Cost Uncrewed Combat Attack System (LUCAS) -- a reverse-engineered combat variant based on a captured Shahed-136 -- which was unveiled at a Pentagon event in July 2025. LUCAS retains the delta-wing planform and pusher-propeller layout but substitutes a two-cylinder DA-215 engine for the four-cylinder Mado, is slightly smaller at approximately 3.0 meters long with a 2.4-meter wingspan, and incorporates an open-architecture modular payload bay accommodating reconnaissance sensors, electronic warfare modules, or explosive warheads. LUCAS is integrated into the Multi-domain Unmanned Systems Communications mesh network, enabling swarm coordination and use as an airborne communications relay. At approximately $35,000 per unit, it is cost-competitive with Iranian and Russian production. CENTCOM deployed LUCAS in combat for the first time on 28 February 2026 during Operation Epic Fury, striking Iranian military targets with a weapon derived from Iran's own design -- a development CENTCOM described explicitly as "delivering American-made retribution" with drones "modeled after Iran's Shahed drones." Griffon Aerospace also developed the MQM-172 Arrowhead, initially as a target drone for training air defenses against the Shahed threat.

China has developed multiple Shahed-inspired systems. The Sunflower-200, produced by Cobtec and first displayed at the Army-2023 exhibition in Russia, is a close copy of the Shahed-136 with a delta-wing layout, pusher propeller, and RATO launch. Its reported specifications include a length of 3.2 meters, a wingspan of 2.5 meters, a maximum takeoff weight of 175 kg, a combat payload of 40 kg, a cruise speed of 160 to 220 km/h, and a flight range of 1,500 to 2,000 km. The ZT-180 is a related Chinese design with a 2.8-meter length, 2.6-meter wingspan, cruise speed of approximately 210 km/h, endurance of six hours at 40 kg payload, a maximum takeoff weight of 180 kg, and an operating altitude of up to

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Poland developed the PLargonia, a scaled-down variant with a reported range of approximately 600 km, which Ukraine has used to strike Russian energy infrastructure. Ukraine has also developed its own derivative drone, sometimes designated the Batyar, adapted from captured or downed examples. Saudi Arabia has reportedly fielded the UnmannedX X-1500 in a similar role. Similar development programs are assessed to be underway in North Korea, Turkey, India, and France, reflecting the degree to which the Shahed-136 has established itself as the reference design for affordable long-range strike capability accessible to states without large aerospace industries.

Strategic Implications

The Shahed-136 and its derivatives have altered the cost calculus of air defense in ways that preoccupy military planners across NATO and allied militaries. The economic asymmetry between attack and defense -- a drone costing tens of thousands of dollars compelling a response from interceptors costing millions -- is not resolved by improving intercept rates; it is compounded by them, because each successful intercept generates costs for the defender that the attacker did not bear. Sustained large-scale employment, as demonstrated in Ukraine from 2022 onward and in the Middle East from 2024, depletes interceptor stocks faster than allied production lines can replenish them. The 12-day Israel-Iran war of 2025 and the subsequent March 2026 conflict strained Patriot and THAAD interceptor inventories that had already been drawn down by years of support to Ukraine.

The proliferation of the design to at least six countries within approximately three years of the Shahed-136's first major combat use represented an unusually rapid spread of a militarily significant weapon concept. The enabling factors -- a simple airframe requiring no exotic materials, a commercial piston engine, civilian-grade electronics, and a rudimentary guidance package -- make the design accessible to states and non-state actors alike. Houthi forces in Yemen, Hezbollah in Lebanon, and Iraqi Shia militia groups have all employed Shahed-family drones supplied through IRGC channels against targets across the Middle East, demonstrating that the weapon has effectively become a tool of Iranian regional power projection beyond Iran's own borders.

During a visit to Qatar in May 2025, US President Donald Trump publicly acknowledged the strategic significance of the Iranian design, describing the drones as "very good, fast, and deadly" and noting that when he asked American arms manufacturers to produce a comparable drone, the initial quoted price was $41 million -- against a Shahed unit cost of $35,000 to $40,000. His remarks publicly credited Iran with having established a cost-efficient production model that the United States was compelled to replicate. The subsequent deployment of LUCAS against Iran in March 2026 confirmed that this replication had been achieved, though the broader question of whether the United States could produce sufficient quantities to alter the strategic balance remained the subject of active debate, with the Pentagon's Drone Dominance program targeting 300,000 low-cost drones beginning in early 2026 as its answer.